PRODUCT REQUIREMENTS DOCUMENT

High-End Architectural 3D Rendering & Virtual Staging Platform

A client and production-management platform for photorealistic CGI rendering, manual virtual staging and virtual walkthrough services.

DOCUMENT FIELD	DETAIL
Document version	1.0
Status	Initial product specification
Product type	MERN full-stack web application
Industry	Architecture, real estate and visualization services
Primary customers	Real estate developers and architects
Prepared	11 November 2026

Project definition

A professional web platform through which real estate developers and architects can submit property and architectural materials, request CGI rendering, virtual staging and virtual walkthrough services, track production, review deliverables, request revisions and download approved outputs.

Scope boundary

The website manages the customer, project and delivery workflow. Professional artists create the visual outputs using external 3D design and rendering tools. The platform is not itself a CAD authoring suite or a custom photorealistic rendering engine.

1. Executive Summary

The product is a web-based service platform for architectural visualization. It connects clients who need high-quality property visuals with an internal visualization team that produces those deliverables.

The platform supports three service categories:

- **Architectural CGI rendering:** photorealistic interior and exterior images created from architectural plans, reference materials or supported 3D models.
- **Virtual staging:** professional designers digitally furnish and decorate photographs or visual scenes of empty properties.
- **Virtual walkthroughs:** guided or interactive ways to explore a property, delivered as supported 360-degree scenes, browser-viewable 3D experiences or walkthrough videos.

The first release should prioritize reliable project submission, file handling, team assignment, progress tracking, review, revision and final delivery. Advanced browser-based 3D viewing should be introduced only for formats and workflows the team can support reliably.

1.1 Product goals

- Provide one organized place for clients and the visualization team to manage each project.
- Make project requirements, uploaded source files, feedback and final deliverables traceable.
- Reduce scattered communication and unclear revision history.
- Give clients visibility into project status and pending actions.
- Support a professional presentation and download experience for completed visuals.

1.2 Non-goals

- Building a new photorealistic rendering engine.
- Building a full CAD or BIM authoring application in the browser.
- Automatically generating CGI images or staged rooms.
- Training or integrating AI, machine-learning or generative-image systems.
- Supporting every CAD, BIM or 3D file format without prior technical validation.
- Real-time collaborative editing of 3D scenes.

2. Problem Statement and Proposed Solution

2.1 Problem

Developers and architects need to communicate the appearance and experience of properties that may still be at the design, construction or marketing stage. Floor plans and technical drawings are not always easy for buyers and other stakeholders to interpret. Visualization work also involves source files, design references, deadlines, multiple deliverables and revision cycles.

When these items are exchanged through disconnected email threads, messaging apps and file-sharing links, it becomes difficult to identify the latest brief, current owner, approved version or outstanding client action.

2.2 Proposed solution

Create a centralized platform where a client can create a project, choose one or more visualization services, submit requirements and files, and follow the work through completion. Internal staff can review the brief, request missing information, assign artists, upload work-in-progress or final outputs, and manage revisions. Clients can comment, request changes, approve deliverables and download the final files.

2.3 Value proposition

Turn architectural concepts and property spaces into clear visual experiences, while making the production and approval process easier to manage.

3. Target Users and Roles

User / role	Primary needs	Core permissions
Real estate developer / client	Show upcoming residential or commercial developments; track work; review and download outputs.	Create and view own organization projects; upload files; comment; request revisions; approve deliverables.
Architect / client	Present building concepts and design intent through realistic visuals.	Create and manage own projects; provide drawings and references; review outputs.
Visualization artist	Understand assigned briefs; access source files; upload previews and final outputs.	View assigned projects; update production tasks; upload deliverables; respond to feedback.
Project manager	Triage requests, clarify briefs, plan work and coordinate artists.	Review submissions; request information; assign artists; update statuses; coordinate reviews.
Administrator	Manage users, access, projects and platform configuration.	Manage accounts and roles; access all projects; audit activity; administer platform settings.

A single client account may represent a developer, architecture practice or other business. Organization-level multi-user access can be included if required by the project owner; it should not delay the first working MVP unless explicitly required.

4. Product Scope and Modules

4.1 Authentication and account management

- User registration and login.
- Secure password hashing and password reset.
- JWT-based authentication with protected API routes.
- Role-based access control (RBAC).
- Profile management and account status controls.

4.2 Client dashboard

- Summary of active, completed and awaiting-action projects.
- Project status and latest activity.
- Pending clarification, review or approval actions.
- Recent deliverables and notifications.
- Links to project details and uploaded files.

4.3 Project and service request management

- Create a project with name, description, property type, location (optional), deadline and project references.
- Select one or more service types: CGI rendering, virtual staging and/or virtual walkthrough.
- Capture service-specific requirements, desired output, style references, camera views and notes.
- Save a draft before submission.
- Allow project managers to request missing information before accepting a brief.

4.4 Architectural CGI rendering

- Accept architectural drawings, reference images and supported 3D source files.
- Capture whether the request is for interior, exterior, aerial, day, night or other agreed views.
- Capture materials, lighting, camera-angle and output-resolution preferences.
- Track each rendering request and its production status.
- Allow artists to upload preview and approved high-resolution render files.

Rendering is performed outside the MERN application using professional production software. The platform stores request information and manages the resulting files; it does not calculate photorealistic renders.

4.5 Manual virtual staging

- Allow clients to upload photographs or supported source materials of empty or sparsely furnished rooms.
- Capture room type, preferred design style, furniture requirements, colour palette and reference images.
- Support style labels such as modern, minimalist, luxury, contemporary, Scandinavian, industrial and traditional.
- Assign the request to a visualization artist.
- Allow the artist to upload staged versions and retain the original source image.
- Provide a before-and-after comparison in the client review interface.

Virtual staging is a human-led design and visualization service. Furniture, decor and other interior elements are added by professional designers using appropriate design and rendering tools.

4.6 Virtual walkthroughs

- Allow a client to request a walkthrough and provide supported 3D source files, scene files or reference materials.
- Track walkthrough type and expected output: video, 360-degree/panorama tour or interactive 3D experience.
- Allow the team to upload a video or provide a supported hosted walkthrough link.
- Provide an in-platform playback or viewing page where the output format allows it.
- Treat interactive 3D model viewing as format-dependent and validate supported formats before promising it.

4.7 File and asset management

- Store metadata for source files, reference images, previews and final deliverables.
- Keep files associated with the correct project and service request.
- Record uploader, upload date, file type, file size and version.
- Use private storage and controlled download access.
- Validate file extension, MIME type and size; use configurable limits.
- Support large-file upload strategies where required by the selected storage provider.

4.8 Review, revisions and approval

- Clients can preview deliverables and submit structured feedback.
- A revision request must identify the affected deliverable and describe the requested change.
- Maintain a version history rather than overwriting previously submitted deliverables.
- Allow artists or project managers to respond to revision requests.
- Allow clients to approve the relevant deliverable or mark the project complete after all required outputs are approved.

4.9 Notifications and activity history

- Notify users about project submission, assignment, status changes, new deliverables, revision responses and approval requests.
- Provide in-app notifications in the MVP.

- Email notifications may be added where the delivery environment and account configuration support them.
- Maintain an activity log for important project events.

4.10 Administration

- Manage client and staff accounts and roles.
- View and filter all projects.
- Review submitted briefs and request clarification.
- Assign or reassign project managers and visualization artists.
- Monitor project status, deadlines, revisions and pending approvals.
- Manage service categories and supported file types.

5. Main User Workflows

5.1 Client workflow

- Client registers or signs in.
- Client creates a project and enters property and contact details.
- Client selects CGI rendering, virtual staging, virtual walkthrough or a combination.
- Client completes service-specific requirements and uploads source files and references.
- Client saves a draft or submits the request.
- Project manager reviews the brief and requests clarification if necessary.
- Project is accepted and assigned to the appropriate team members.
- Client tracks status and receives notifications.
- Visualization team uploads a preview or deliverable.
- Client reviews the output, comments, requests revisions or approves it.
- Final approved deliverables are made available for secure download.

5.2 Visualization team workflow

- Team member opens the assigned project and reads the brief.
- Team member accesses permitted source files and references.
- Team member carries out the work in external professional software.
- Team member uploads a preview or deliverable with a version label and notes.
- Project manager performs any required internal review.
- Client receives the review request and provides feedback or approval.
- The artist completes requested revisions and uploads a new version.
- Approved outputs are marked final and made available to the client.

5.3 Project status lifecycle

Recommended project statuses:

Status	Meaning
Draft	Client has started a project but has not submitted it.
Submitted	Client has submitted the request for review.

Status	Meaning
Under Review	Project manager is checking the brief and files.
Awaiting Information	Client must provide clarification or missing materials.
Assigned	Project has an owner or production team.
In Progress	Production work is underway.
Internal Review	Team is checking the output before client review.
Client Review	Deliverable is waiting for client feedback.
Revision Requested	Client has requested changes.
Approved	Client has approved the required output.
Completed	All required deliverables are approved and delivered.
Cancelled	Project has been cancelled by an authorized user.

Status transitions should be validated by the backend. Only authorized roles should perform each transition.

6. Functional Requirements

ID	Area	Requirement	Priority
FR-01	Authentication	Users can register, sign in, sign out and reset passwords.	High
FR-02	Authorization	The API enforces role-based access and project ownership.	High
FR-03	Client dashboard	Clients can see project summaries, statuses and pending actions.	High
FR-04	Project creation	Clients can create, edit and submit project records.	High
FR-05	Service selection	A project can include CGI rendering, virtual staging and/or walkthrough requests.	High
FR-06	Brief capture	The system stores service-specific requirements and references.	High
FR-07	File upload	Authorized users can upload source files and references to private storage.	High
FR-08	Project review	Project managers can review submissions and request missing information.	High
FR-09	Assignment	Authorized staff can assign and reassign projects to team members.	High
FR-10	Status tracking	Authorized users can update valid project statuses; clients can track progress.	High
FR-11	Deliverables	Artists can upload previews and final images, videos or supported 3D assets.	High
FR-12	Review and comments	Clients and staff can comment on deliverables and project activity.	High
FR-13	Revision workflow	Clients can request revisions against a specific deliverable version.	High
FR-14	Approval	Clients can approve deliverables; approval events are recorded.	High
FR-15	Secure download	Clients can download only files they are authorized to access.	High
FR-16	Notifications	The system creates in-app notifications for key project events.	Medium
FR-17	Activity history	The system records important actions and status changes.	Medium

ID	Area	Requirement	Priority
FR-18	Before/after view	Virtual staging outputs can be compared with their original images.	Medium
FR-19	Walkthrough playback	The platform can play supported uploaded walkthrough videos or open supported hosted tours.	Medium
FR-20	Interactive 3D viewer	The platform can display selected supported 3D formats in-browser after technical validation.	Future / conditional
FR-21	Reporting	Admins can view basic project counts, status distribution and overdue work.	Future
FR-22	Payments and subscriptions	Online billing and subscription plans may be considered after business requirements are confirmed.	Future

7. Non-Functional Requirements

7.1 Security

- Hash passwords using a modern password-hashing algorithm.
- Enforce authorization on the server for every project and file operation.
- Do not expose permanent public URLs for private source files.
- Validate uploads using extension, MIME type, size and storage policy.
- Use secure, expiring links or authenticated download endpoints.
- Protect against common web risks including injection, broken access control, CSRF where applicable and unsafe file uploads.
- Keep secrets and credentials in environment variables or a secrets manager.

7.2 Performance and reliability

- Paginate project lists, activity feeds and deliverable lists.
- Avoid loading large images, videos or 3D files as part of ordinary dashboard API responses.
- Use thumbnails and optimized previews where practical.
- Handle large uploads and downloads without blocking normal API requests.
- Provide clear error messages and retry-safe upload behaviour.
- Record operational errors without logging passwords, tokens or private file contents.

7.3 Scalability and maintainability

- Organize the backend into modular routes, controllers, services, models and middleware.
- Separate file-storage operations from business logic.
- Use indexes for common queries such as project owner, status, assignee and creation date.
- Keep rendering production external so the web platform can scale independently from artists' workstations and rendering tools.
- Document API contracts and environment configuration.

7.4 Compatibility and accessibility

- Support current desktop versions of Chrome, Edge, Firefox and Safari.
- Make core client and staff workflows usable on mobile browsers.
- Use accessible labels, keyboard navigation and readable contrast for primary workflows.

• Treat advanced 3D viewing as a capability that may require more powerful devices and browsers.

8. Proposed Technology Stack

Layer	Proposed technology	Responsibility
Frontend	React.js, JavaScript, React Router, Axios, Tailwind CSS	Client portal, staff dashboard, admin views, project forms and review screens.
Backend	Node.js, Express.js	REST APIs, business rules, authentication, permissions and workflows.
Database	MongoDB, Mongoose	Users, organizations, projects, service requests, assignments, assets and feedback.
Authentication	JWT, secure password hashing	Session/token-based API access and role checks.
Real-time updates	Socket.IO (optional for MVP)	Live status and notification updates where useful.
File storage	Cloudinary for image assets; S3-compatible object storage for large files	Private source files, previews, videos and final deliverables.
3D browser viewing	Three.js / React Three Fiber / WebGL	Optional browser viewer for specifically supported 3D formats.
Production software	Blender, D5 Render, Unreal Engine or other tools selected by the visualization team	Professional modeling, staging, rendering, animation and walkthrough production.

Technology choices are proposals, not a mandate to integrate every listed tool in the first release. The team should confirm hosting, storage, file formats and production tools with the project owner.

9. High-Level System Architecture

The MERN application coordinates the workflow. External production tools are used by artists to create the visual outputs.

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CLIENT / STAFF BROWSER
React UI: dashboards • project forms • uploads • review • deliverable viewer

↓ HTTPS / REST API (and optional Socket.IO)

NODE.JS + EXPRESS BACKEND
Authentication • RBAC • project workflow • assignments • revisions • notifications

■ ↓ ■
MONGODB OBJECT STORAGE NOTIFICATION SERVICE
Users / projects Source files / renders In-app / email (if enabled)

EXTERNAL VISUALIZATION WORKFLOW
Artists use Blender / D5 Render / Unreal Engine or other approved tools
↓ Upload previews, final CGI images, staged images, videos or supported tour assets

CLIENT REVIEW AND SECURE DELIVERY
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The application should not attempt to run heavy CGI rendering inside ordinary API requests. Rendering jobs are production work items managed by the platform, not compute jobs executed by the MERN server.

9.1 Key backend responsibilities

• Authenticate users and enforce access permissions.

• Validate project and service request data.

• Create and manage assignments and status transitions.

• Generate secure upload/download authorization for storage.

• Store deliverable metadata and version history.

- Record comments, revision requests, approvals and activity events.
- Trigger notifications for relevant workflow changes.

10. Initial MongoDB Data Model

Collection	Purpose / key fields
users	name, email, passwordHash, role, organizationId, accountStatus, createdAt
organizations (if multi-user clients are required)	name, organizationType, contactDetails, memberIds
projects	projectCode, name, description, propertyType, location, clientId/organizationId, status, deadline, createdAt
serviceRequests	projectId, serviceType, requirements, stylePreferences, outputSpecifications, status
assets	projectId, serviceRequestId, uploadedBy, storageKey, originalName, MIME type, size, category, version, visibility
assignments	projectId, serviceRequestId, assignedTo, assignedBy, roleOnProject, assignedAt
deliverables	projectId, serviceRequestId, assetId, deliverableType, version, notes, uploadedBy, reviewStatus
revisionRequests	deliverableId, requestedBy, description, status, response, createdAt
comments	projectId, deliverableId (optional), authorId, message, createdAt
notifications	recipientId, type, message, relatedProjectId, readAt, createdAt
activityLogs	actorId, projectId, action, entityType, entityId, timestamp, metadata

Use MongoDB references for entities that need independent access, querying or lifecycle management. Avoid duplicating large file binaries in MongoDB; store files in object storage and retain metadata and storage keys in the database.

10.1 Data integrity rules

- Every project must have an owning client or organization.
- Every service request must belong to a project.
- Every deliverable must belong to the relevant project and, where applicable, service request.
- A revision request must reference a specific deliverable version.
- Only authorized users may access project files and perform status transitions.
- Approval must record the approving user and timestamp.

11. Competitor Overview

The following is a high-level landscape based on publicly available product and service descriptions. These products are reference points, not a feature checklist. Their business models differ: some are visualization services, some are design/rendering software, and some focus on scanning existing spaces.

Competitor	Relevant public offering	What it means for this product
BoxBrownie	Human-produced virtual staging, property image editing, architectural renders and 360° render tours.	Closest service-workflow reference: customers submit source materials and receive finished visual marketing assets.
Coohom	Interior design tools, furniture/material libraries, image/video rendering and 360° walkthrough outputs.	Useful reference for how design outputs and 360° presentations can be delivered; it is also a design platform.

Competitor	Relevant public offering	What it means for this product
Cedreo	Browser-based home design, floor plans, furnishing and photorealistic 3D renderings.	A software-led alternative for users who want to create designs themselves, rather than only submit production requests.
D5 Render	Professional real-time architectural rendering, animation, scene presentation and panorama/spatial tours.	A production tool reference for the internal visualization team, not a direct replacement for the client-management portal.
Matterport	3D capture/digital twins, immersive tours, measurements, guided tours and downloadable assets.	A reference for property walkthrough experiences, especially for existing spaces captured through 3D scanning.

Differentiation opportunity: deliver a focused client-to-artist workflow for developers and architects that combines CGI requests, manual virtual staging, walkthrough deliverables, project tracking, revisions and approvals in one place. This is a product hypothesis to validate with the project owner and prospective users, not a claim that no competitor offers similar features.

11.1 Official competitor references

• BoxBrownie – Virtual staging and property visualization services
https://www.boxbrownie.com/services/virtual-staging

• BoxBrownie – 360° render virtual tours
https://dev4.boxbrownie.com/renders-360-virtual-tours

• Coohom – 3D interior design and rendering
https://www.coohom.com/enterprise/floor-planner

• Cedreo – 3D home design features
https://cedreo.com/features/

• D5 Render – Features
https://www.d5render.com/features

• Matterport – 3D camera and virtual tour platform
https://go.matterport.com/PS_EN.html

12. MVP Definition and Delivery Roadmap

The MVP should prove that a client can submit a complete visualization request and receive an approved deliverable through a managed workflow. It should not attempt to reproduce the full feature set of professional 3D design software.

Phase	Scope	Exit criteria
Phase 1 – Foundation	Repository setup, React app, Express API, MongoDB, environment configuration, authentication and RBAC.	Users can sign in and access role-appropriate areas.
Phase 2 – Project intake	Client dashboard, project creation, service selection, brief forms, file uploads and submission.	A client can submit a complete project with source files and requirements.
Phase 3 – Team operations	Admin/project manager views, brief review, clarification, assignment and status tracking.	Staff can review, assign and track submitted projects.
Phase 4 – Production and delivery	Artist workspace, preview/final uploads, deliverable versions, secure preview and download.	Assigned staff can upload outputs and clients can access authorized deliverables.
Phase 5 – Review and revisions	Comments, revision requests, approval actions, notifications and activity history.	A project can complete a full review/revision/approval cycle.
Phase 6 – Walkthrough presentation	Video playback and/or supported hosted tour links; evaluate browser 3D viewer for selected formats.	Clients can open supported walkthrough deliverables; limitations are documented.

12.1 Recommended MVP priorities

- Must have: authentication, RBAC, project creation, service request forms, file upload, assignment, status tracking, deliverable upload, revisions, approval and secure download.
- Should have: in-app notifications, activity history, before/after staging comparison and walkthrough video/link viewing.
- Defer until validated: advanced interactive 3D viewer, complex team collaboration, payments, subscriptions and detailed analytics.

13. Acceptance Criteria and Success Measures

13.1 MVP acceptance criteria

- A client can register, sign in and create a project.
- A client can select CGI rendering, virtual staging or walkthrough services and provide the required brief information.
- A client can upload source files and references, and those files remain associated with the correct project.
- A project manager can review the submission, request clarification and assign work.
- An assigned artist can access the authorized source materials and upload a preview or deliverable.
- A client can review a deliverable, request a revision or approve it.
- The system retains prior deliverable versions and records revision/approval history.
- A client can securely download approved final deliverables.
- Users cannot access projects or files outside their permissions.
- The project can be completed without any AI feature or custom rendering engine.

13.2 Operational measures to track

- Number of projects submitted, active and completed.
- Average time from submission to assignment.
- Average time spent in each project status.
- Number of revision cycles per deliverable.
- Percentage of projects completed by their agreed deadline.
- Number of projects awaiting client information or approval.
- Upload failures and download errors.

Targets for these measures should be set after the organization establishes expected service levels and observes pilot projects. This PRD does not invent turnaround-time or quality guarantees.

14. Risks, Dependencies and Open Decisions

Topic	Risk / decision needed
Source file formats	Confirm which drawing and 3D formats the production team actually accepts. Do not promise universal CAD/BIM compatibility.
Large files	Confirm typical and maximum file sizes, storage provider, retention policy and upload method.
Walkthrough delivery	Decide whether the initial service produces video, 360° tours, interactive 3D scenes or a combination.
Browser 3D viewer	Validate formats, conversion requirements, device performance and hosting before committing to an in-browser viewer.
Production ownership	Confirm who creates CGI renders and staged images, who performs quality review and who communicates with clients.

Topic	Risk / decision needed
Revision policy	Define included revision rounds, what counts as a revision and how scope changes are handled.
Deadlines and pricing	Confirm quotation, payment, cancellation and turnaround rules separately; these are not assumed in the MVP.
Client organizations	Confirm whether multiple users from one developer or architecture firm need shared project access.

15. Explicit Out-of-Scope Items

- AI-powered virtual staging, generative image editing, AI recommendations or AI assistants.
- A custom photorealistic CGI rendering engine.
- Full CAD/BIM editing and authoring in the browser.
- Guaranteed import and conversion of every 2D/3D design format.
- Real-time multi-user editing of 3D scenes.
- Automatic model reconstruction from ordinary photographs.
- Payment gateway, subscription and invoicing automation until business rules are confirmed.
- Native mobile applications; the initial product is a responsive web application.

16. Assumptions and Constraints

- The initial product is a service-management platform for professional visualization work, not a self-service CAD product.
- Clients provide source drawings, photos, models and design references appropriate to the requested service.
- The internal visualization team uses external professional tools to produce outputs.
- The platform stores and presents deliverables; it does not execute the heavy rendering workload.
- The first release is a responsive MERN web application.
- Only explicitly tested file types and walkthrough formats will be advertised as supported.
- Final scope, hosting, storage limits, team structure and service policies require confirmation from the project owner.

17. Final Product Definition

The High-End Architectural 3D Rendering & Virtual Staging Platform is a MERN-based web application for real estate developers and architects to request and manage professional architectural visualization services. It supports photorealistic CGI rendering, manually produced virtual staging and virtual walkthrough deliverables through a structured workflow covering project intake, file management, team assignment, production tracking, review, revisions, approval and secure delivery.

The platform's value is not that it replaces professional 3D tools. Its value is that it makes the end-to-end client and production workflow organized, visible and dependable.

Document status: Initial PRD v1.0. Requirements should be validated with the project owner and visualization team before implementation begins.